Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 34488

Part A: Each question is worth 3 points

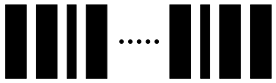
1. A street has 126 houses, numbered from 1 to 126. Leo writes all the house numbers on cards, one digit on each card.
How many cards does Leo need?

A. 261 ***B.*** 270 ***C.*** 279 ***D.*** 322 ***E.*** 378

2. Jumpy the kangaroo's longest jump in training was 60 dm 20 cm and 30 mm long. At the competition the jump was 159 cm longer.
How long was the jump at the competition?

A. 2 m 42 cm ***B.*** 7 m 82 cm ***C.*** 8 m 9 cm ***D.*** 8 m 82 cm ***E.*** 22 m 13 cm

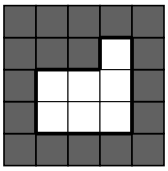
3. A bar code is made of 29 black and white bars, black and white taking turns. The first and the last bar are black. Each black bar is either wide or narrow. There are 5 more white bars than wide black bars.

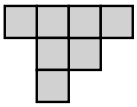
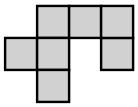
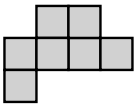
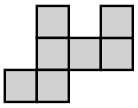
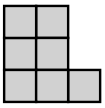


How many narrow black bars are there?

A. 10 ***B.*** 9 ***C.*** 7 ***D.*** 6 ***E.*** 5

4. Which piece covers the empty area exactly? The pieces may be turned and flipped over.



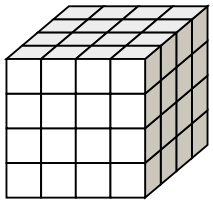
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part B: Each question is worth 4 points

5. Omar threads 19 beads on a string: red and blue take turns, and the string starts and ends with a red bead. Each red bead is big or small. There are 3 more blue beads than big red beads.
How many big red beads are there?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 4

6. Chris glued 64 small cubes into a big 4 × 4 × 4 cube (like the picture) and painted the whole outside of it, also the bottom. Then the big cube falls apart.



How many small cubes have paint on exactly three faces?

A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

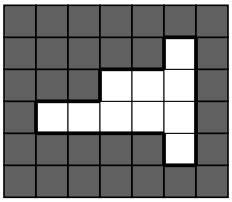
7. Leo builds a brick 5 cubes long and 5 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Leo used exactly 18 blue cubes. How many cubes deep is the brick?

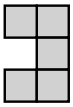
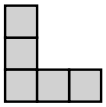
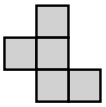
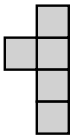
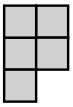
A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

8. Eva has a rope 3 m long and cuts it into pieces that are each 25 cm long. How many pieces does Eva get?

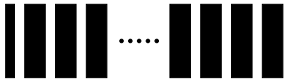
A. 12 ***B.*** 11 ***C.*** 10 ***D.*** 9 ***E.*** 8

9. Two pieces that are exactly the same cover the empty area. Which piece is it? The pieces may be turned and flipped over.



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

10. A bar code is made of 23 black and white bars taking turns, with a black bar at each end. Each black bar is wide or narrow. There are 1 more white bars than wide black bars.



How many wide black bars are there?

A. 12 ***B.*** 11 ***C.*** 10 ***D.*** 9 ***E.*** 2

11. Sara reads a chapter from page 79 to page 138 (both included).
How many digits are printed in the page numbers of this chapter?

A. 60 ***B.*** 118 ***C.*** 157 ***D.*** 159 ***E.*** 161

12. Nina threads 19 beads on a string: red and blue take turns, and the string starts and ends with a red bead. Each red bead is big or small. There are 1 more blue beads than big red beads.
How many small red beads are there?

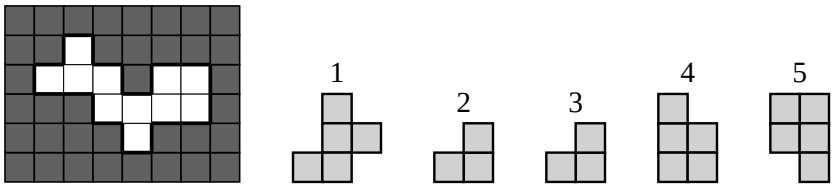
A. 8 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

13. Eva's snake toy is 12 dm 9 cm long. Finn's snake toy is 920 mm long.
How many centimetres longer is Eva's snake?

A. 37 cm ***B.*** 46 cm ***C.*** 47 cm ***D.*** 71 cm ***E.*** 120 cm

Part C: Each question is worth 5 points

14. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+2+4 ***C.*** 1+2+5 ***D.*** 1+4+5 ***E.*** 3+4+5

15. Exactly 405 digits are used to write all the page numbers in a book, starting from 1.
How many pages are there?

A. 171 ***B.*** 170 ***C.*** 135 ***D.*** 81 ***E.*** 72

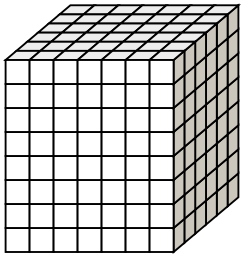
16. Cleo's new rug is 3 m 9 cm long. Four of these lengths are the same as that.
Which one is **not** the same?

A. 309 mm ***B.*** 3090 mm ***C.*** 30 dm 9 cm ***D.*** 309 cm ***E.*** 3 m 90 mm

17. A bar code is made of 14 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 2 more white bars than wide black bars.
How many narrow black bars are there?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

18. Sara builds a brick 7 cubes long, 6 cubes deep and 8 cubes tall from red and blue cubes. All the cubes you can see from the outside are red. The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes. All the other cubes, in the middle, are blue.



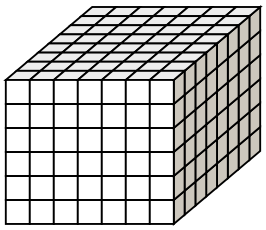
How many blue cubes are there?

A. 6 ***B.*** 24 ***C.*** 48 ***D.*** 60 ***E.*** 312

19. Exactly 261 digits are used to write all the house numbers in a street, starting from 1. How many houses are there?

A. 24 ***B.*** 33 ***C.*** 87 ***D.*** 122 ***E.*** 123

20. Mia builds a brick 7 cubes long, 8 cubes deep and 6 cubes tall from red and blue cubes. All the cubes you can see from the outside are red. The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes. All the other cubes, in the middle, are blue.



How many blue cubes are there?

A. 312 ***B.*** 120 ***C.*** 60 ***D.*** 48 ***E.*** 24

Answer Key

1.	**B**	**2.**	**B**	**3.**	**D**	**4.**	**E**	**5.**	**D**	**6.**	**E**
7.	**E**	**8.**	**A**	**9.**	**D**	**10.**	**C**	**11.**	**D**	**12.**	**C**
13.	**A**	**14.**	**A**	**15.**	**A**	**16.**	**A**	**17.**	**C**	**18.**	**B**
19.	**E**	**20.**	**E**								

Solutions

1. B (270) — 9 × 1 + 90 × 2 + 27 × 3 = 270. [set-03/by_level/Grade_03-04/2003 Q16 · digits_for_pages]

2. B (7 m 82 cm) — 60 dm = 6 m and 30 mm = 3 cm, so the training jump was 6 m 23 cm. Adding 159 cm gives 7 m 82 cm. [set-03/by_level/Grade_03-04/2003 Q10 · jump]

3. D (6) — Black is at both ends, so there are 15 black and 14 white bars. Wide: 14 – 5 = 9. Narrow: 15 – 9 = 6. [set-03/by_level/Grade_03-04/2003 Q17 · narrow_bars]

4. E — The empty area has 7 squares; all pieces do too, so look at the shape. Turn piece E and it fits exactly; each of the others has one square in a different place. [set-03/by_level/Grade_03-04/2003 Q18 · one_piece]

5. D (6) — There are 10 red and 9 blue beads. Big red: 9 – 3 = 6. [set-03/by_level/Grade_03-04/2003 Q17 · necklace]

6. E (8) — They are the corners, and a cube always has 8 corners. [set-03/by_level/Grade_03-04/2003 Q9 · painted]

7. E (4) — The blue inside is 3 long and 3 tall, so one layer of it has 3 × 3 = 9 cubes. 18 ÷ 9 = 2 layers of blue, plus the red front and back: 4. [set-03/by_level/Grade_03-04/2003 Q9 · how_deep]

8. A (12) — 3 m = 300 cm. 300 ÷ 25 = 12. [set-03/by_level/Grade_03-04/2003 Q10 · pieces]

9. D — The empty area has 10 squares, so each piece has 5; every option has 5. Only two copies of piece D fit together into the shape of the hole. [set-03/by_level/Grade_03-04/2003 Q18 · two_copies]

10. C (10) — There are 12 black and 11 white bars. Wide: 11 – 1 = 10. [set-03/by_level/Grade_03-04/2003 Q17 · wide_bars]

11. D (159) — There are 60 pages: 21 × 2 + 39 × 3 = 159 digits. [set-03/by_level/Grade_03-04/2003 Q16 · from_page]

12. C (2) — There are 10 red and 9 blue beads. Big red: 9 – 1 = 8. Small red: 10 – 8 = 2. [set-03/by_level/Grade_03-04/2003 Q17 · necklace]

13. A (37 cm) — 12 dm 9 cm = 129 cm and 920 mm = 92 cm. 129 – 92 = 37 cm. [set-03/by_level/Grade_03-04/2003 Q10 · how_much_longer]

14. A (1+2+3) — The empty area has 11 squares, so the three sizes must add up to 11. Only 1 + 2 + 3 also fit together in the right shape. [set-03/by_level/Grade_03-04/2003 Q18 · three_pieces]

15. A (171) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 405 – 189 = 216 digits are three-digit numbers: 216 ÷ 3 = 72, from 100 to 171. [set-03/by_level/Grade_03-04/2003 Q16 · three_digit_pages]

16. A (309 mm) — 3 m 9 cm = 309 cm = 3090 mm. But 309 mm = 309 mm. [set-03/by_level/Grade_03-04/2003 Q10 · not_equal]

17. C (2) — Starting black and ending white, the bars come in pairs: 7 black and 7 white. Wide: 7 – 2 = 5. Narrow: 7 – 5 = 2. [set-03/by_level/Grade_03-04/2003 Q17 · ends_differ]

18. B (24) — Each length loses two red cubes at each end, so it drops by 4: 3 × 2 × 4 = 24. [set-03/by_level/Grade_03-04/2003 Q9 · thick_wall]

19. E (123) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 261 – 189 = 72 digits are three-digit numbers: 72 ÷ 3 = 24, from 100 to 123. [set-03/by_level/Grade_03-04/2003 Q16 · three_digit_pages]

20. E (24) — Each length loses two red cubes at each end, so it drops by 4: 3 × 4 × 2 = 24. [set-03/by_level/Grade_03-04/2003 Q9 · thick_wall]